Territorial Information Retrieval from Heterogeneous Open Data through the Construction of a Data Warehouse for Water Management in Mexico City

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Abstract. This work presents a data warehouse for the territorial information retrieval of water consumption in Mexico City (CDMX), built from heterogeneous open sources. The contribution lies in three methodological elements: an ETL process that consolidates official SACMEX data into a unified dimensional model, a declarative R2RML mapping that republishes the resulting star schema as an RDF knowledge graph whose Data Cube structure is validated against the vocabulary's integrity constraints, so that territorial retrieval can also be expressed as graph traversal, and a deployment methodology that guarantees an accessible demonstration through a static version published on GitHub Pages and backed by an updated public repository. The prototype exposes the model through a retrieval interface with an interactive choropleth map, filters by year, bimester, borough (*alcaldía*) and neighborhood (*colonia*), and an exploratory module for atypical consumption, turning dispersed open data into queryable information for urban diagnosis, research and citizen monitoring.

Keywords: Data warehouse · Information retrieval · Open data · ETL · Knowledge graph · RDF Data Cube · Water consumption · Mexico City

1 Introduction

Mexico City faces a water crisis of aquifer overexploitation, network leaks and population growth. Open data on water consumption exist (SACMEX), but in

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poorly accessible formats and without query tools, which limits their use for evidence-based decision making.

This work addresses the problem from two perspectives proper to information processing: the extraction of structured information from heterogeneous open sources through an ETL process, and the retrieval of that information through a parameterized query interface, transforming dispersed datasets into a queryable dimensional repository that anyone can filter and visualize geospatially by neighborhood and borough.

The contribution of this work is threefold: (i) a reproducible ETL process to consolidate official open data into a dimensional data warehouse, (ii) a declarative R2RML mapping that republishes it as a validated RDF knowledge graph, and (iii) a continuous-deployment methodology that keeps a functional demonstration always accessible, combining a dynamic deployment connected to a database with a static version published in a public repository.

The analysis of the 70,886 consolidated records reveals that the boroughs of Cuauhtémoc and Miguel Hidalgo lead the total consumption (Table 3), and that consumption varies across the available bimesters without a direct causal relationship with dry or rainy periods. The system serves citizens querying their own neighbourhood, public servants diagnosing territorial demand, researchers, journalists and civil organisations monitoring water management.

Section 2 reviews related work, Section 3 describes the methodology, Section 5 presents the results and Section 7 the conclusions.

2 Related Work

2.1 Foundations: Data Warehouses, Dimensional Modeling, and ETL

Dimensional modeling and the star schema constitute the consolidated basis for the design of analytical data warehouses. Kimball and Ross [1] establish the canonical principles —fact and dimension tables, surrogate keys, slowly changing dimensions, and the ETL subsystems— that ground the model adopted in this work. On the integration phase, Vassiliadis [2] offers a reference survey of *Extract–Transform–Load* technology, covering the conceptual and logical modeling of the processes and the cross-cutting extraction, transformation, and loading issues that methodologically support our pipeline.

2.2 Data Warehouses and Information Systems for Water Management

In the water domain, the most direct antecedent is that of Soares et al. [3], who implement a data warehouse with an ETL process to monitor water supply and consumption, integrating meter readings, reservoir volumes, and geographic location from management-application databases and heterogeneous files. Abecker et al. [4] present a hybrid sensor-semantic architecture (the WatERP project) that combines a relational/PostGIS store with a triple store for integrated resource management. These systems, however, mostly start from operational data, SCADA, or proprietary sensor networks, and not from open government data.

2.3 The Gap: ETL over Heterogeneous Open Data and Data Quality

Integrating open government data into dimensional warehouses poses its own challenges. Berro, Megdiche, and Teste [6] propose “content-driven” ETL processes that address precisely the semantic and structural heterogeneity, the absence of schemas, and the dispersion characteristic of open data, factors that overwhelm traditional ETL. Complementarily, Vetrò et al. [5] define a quantitative measurement framework for the quality of Open Government Data (accuracy, completeness, consistency, traceability, and freshness) and empirically report that the most frequent problems are missing metadata, incomplete data, and the absence of update information —exactly the challenges our ETL faces over the SACMEX data.

2.4 Geospatial Information Retrieval and Visualization

Purves et al. [7] offer a reference monograph on Geographic Information Retrieval (GIR) —indexing, search, and spatial ranking— that theoretically frames the territorial information-retrieval component of this work.

2.5 Knowledge Graphs for Urban and Water Domains

Republishing a relational warehouse as RDF is a mature practice with W3C standards for both ends. The RDF Data Cube vocabulary [12] models statistical observations over conformed dimensions, which is the same structure a star schema encodes; R2RML [13] specifies the mapping declaratively, so the graph is derived rather than maintained by hand; and GeoSPARQL 1.1, documented by Car and Homburg [14], supplies the geometry model and the topological vocabulary that a flat territorial dimension cannot express. In the water and urban domains the direction is established: Roussey et al. [15] survey knowledge-graph technologies as the next frontier for food, agriculture and water. What these works share is the premise this article adopts: the graph is valuable for interoperability and topological querying, not as a replacement for the analytical store.

2.6 Reproducibility and Deployment of Scientific Demonstrations

Finally, Sandve et al. [8] formulate widely cited rules for computational reproducibility (version control, documentation of preprocessing, public access to data and scripts) that ground our reproducible static version. Hill et al. [9] provide a practical guide to developing and deploying an open-source, reproducible web dashboard for water-sanitation data, a close domain that supports the continuous-deployment methodology with reproducible artifacts.

2.7 Positioning of the Contribution

Unlike existing water data warehouses —which start from operational data, SCADA, or proprietary sensors [3,4]— and recognizing that the integration of heterogeneous open data is a documented but unsolved problem in this domain [6,5], this work differs in three respects. First, an ETL process designed to consolidate official SACMEX open data into a dimensional star schema. Second, the exposure of the information through parameterized queries and a choropleth geospatial visualization that enables territorial information retrieval [7], published as static artefacts that require no server. Third, a continuous-deployment methodology accompanied by a reproducible static version that guarantees the indefinite verifiability of the system [8,9]. This same approach of consolidating official open data was previously applied to the seismic-risk domain in Mexico [10], a direct methodological antecedent transferred here to the water domain. To the best of our review, no academic, peer-reviewed, and reproducible data warehouse built specifically over SACMEX/CDMX open data exists.

3 Methodology

3.1 General Architecture

The system follows a layered architecture: a data layer (PostgreSQL), a processing layer (ETL with Python and SQL), a query layer of parameterized SQL and, over it, a declarative RDF projection, and a presentation layer (HTML/CSS/JS frontend with Leaflet and Plotly). The warehouse implements a star schema over two fact tables —`fact_consumo_agua`, the subject of this article, and `fact_clima`, the climate series joined through the shared temporal dimension— and three conformed dimensions: `dim_tiempo` (date, year, bimester), `dim_ubicacion` (borough, neighborhood, latitude, longitude) and `dim_indice_des` (development level). Identifiers are quoted verbatim from the published schema.

3.2 ETL Consolidation Process

The first contribution of the work is the ETL process that consolidates the SACMEX open data into the dimensional warehouse. The complete detail of the process —the DDL and the ETL scripts (`ddl/0_schema.sql`, `etl/1_copy.sql`, `etl/2_dim.sql`, `etl/3_fact.sql`) and the parameterized retrieval queries— is documented in the project's public repository; this article presents only the architecture of the process.

The pipeline operates in four phases. (1) **Extraction**: reading the open SACMEX CSV files. (2) **Validation and cleaning**: discarding records with empty fields, inconsistent coordinates, or incompatible formats. (3) **Loading to staging and populating dimensions**: `COPY` of the data into temporary tables and extraction of the unique values into `dim_tiempo`, `dim_ubicacion` and `dim_indice_des`. (4) **Populating the fact tables**: a join between staging and dimensions that populates `fact_consumo_agua` and `fact_clima`, finally dropping the staging tables. The information thus consolidated becomes available for retrieval through the parameterized queries distributed with the archived release.[1]

3.3 Continuous Deployment Methodology

So that the demonstration survives without a server, the system is published in two modalities. The **dynamic deployment** keeps the complete architecture over PostgreSQL, serves parameterized queries and produced the results reported here. The **static version** exports the data as `JSON` and `GeoJSON` that the frontend consumes directly, so querying, filtering, aggregation and visualization run on the client. The duality addresses the recurring problem that academic prototypes stop working shortly after evaluation [8,9]. The system is available at https://github.com/gabrielhuav/Data_Warehouse_static (repository), https://gabrielhuav.github.io/Data_Warehouse_static/ (static demo, no backend required) and https://github.com/gabrielhuav/data_warehouse_cdmx (warehouse, preservation fork).
What is preserved deserves precision. The archived release (https://github.com/gabrielhuav/Data_Warehouse_static/releases/tag/v1.5-icokg2026) distributes the warehouse itself —schema, ETL, both source files and the container definition that rebuilds the database, vendored unmodified from its author's repository and additionally preserved as a fork— together with the static frontend, the R2RML mapping, a browser-side SPARQL explorer (https://gabrielhuav.github.io/Data_Warehouse_static/grafo.html) and the scripts that regenerate the exported data, so the JSON the demonstration consumes is an export of the warehouse and not a separate dataset. Citations refer to the frozen release; the repository head may evolve, the release will not.

3.4 Data Scope, Grain and Cleaning Criteria

Three properties of the source bound every result reported below. *Extent*: the resource is published as *Consumo Agua (1er Semestre 2019)* [11] and contains exactly that —the first three bimesters of 2019, one reading date each, in 71,102 records— so the warehouse is not a multi-year series and the seasonal variation it supports is a within-semester comparison. The bound is declared by the publisher in the title of the resource, not chosen by us, and the portal reports no release after February 2023: a partial extraction published once, without versioning, which is the problem this article studies. Against the source becoming unavailable, both files the ETL consumes are redistributed with the archived release and the portal page is preserved in the Internet Archive. *Grain*: one

[1] SACMEX operates a proprietary platform integrating SCADA and multiple sources; it is distinguished from a reproducible academic work over open data such as this one.

Table 1: Records discarded during validation and cleaning, over the 71,102 records of the source export.

Exclusion criterion	**Records**	**% of source**
Unmatched territorial label	216	0.30
Missing development index	0	0.00
Reading date absent from the time dimension	0	0.00
Total discarded	216	0.30
Loaded into the fact table	70,886	99.70

fact row is one *manzana* (city block) for one billing period, and the export carries 22,930 distinct coordinate-bearing units over 1,553 borough–neighbourhood pairs, so the location dimension is coarser than the grain and every aggregate is a roll-up. *Scale*: the development index has four ordinal levels —ALTO, MEDIO, BAJO and POPULAR— published with the dataset rather than derived here.

The exclusion criteria are deterministic and enforced structurally: the fact table is populated by inner joins against the three dimensions, so a record that fails to resolve a dimension member is not loaded. Table 1 quantifies each criterion rather than merely declaring data quality [5]. The distribution is unusually clean and that is itself a finding: a single criterion accounts for every discarded record —the 216 rows carrying the literal `NA` in both territorial fields— and none is lost to a malformed measure or an unparseable date. This is a property of the SACMEX export rather than a merit of the process.

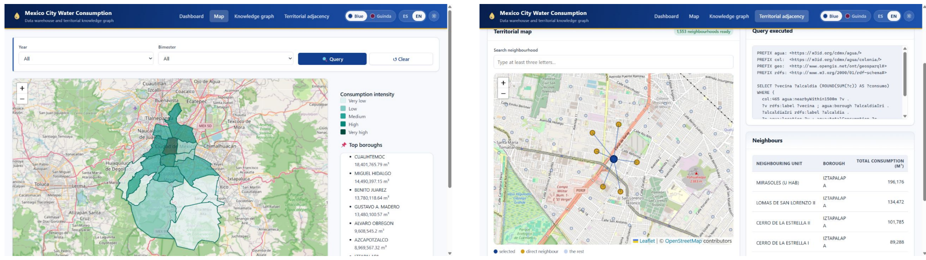
Data sources and attribution. The meteorological series comes from Open-Meteo under CC BY 4.0 and the cartographic base of the interactive map is OpenStreetMap under the Open Database License, which requires attribution wherever the map is reproduced.

3.5 Frontend and Geospatial Visualization

The frontend offers a dashboard with indicators, interactive charts (Plotly), and a choropleth map of the 16 boroughs (Leaflet + GeoJSON), where color represents the consumption level and each borough displays a popup with its detail (Fig. 1a). Alongside it, and over the same exported data, a second view resolves territorial adjacency by traversing the knowledge graph of Section 4 rather than by filtering the cube: selecting a neighbourhood returns its neighbours ranked by consumption, together with the `agua:nearbyWithin1500m` query that produced them (Fig. 1b). The two views in Fig. 1 are therefore the same warehouse read through its two representations, and both run in the browser with no server behind them.

3.6 Implementation of the Atypical-Consumption Module

The current version replaces supervised prediction with an exploratory detection module oriented to analytical retrieval: because the open source carries no



(a) Choropleth retrieval by borough. https://gabrielhuav.github.io/Data_Warehouse_static/mapa.html

(b) Adjacency as graph traversal, with the `agua:nearbyWithin1500m` query and its result. https://gabrielhuav.github.io/Data_Warehouse_static/vecindad.html?alcaldia=IZTAPALAPA&colonia=A0%20DE%20JUAREZ

Fig. 1: The two retrieval modes of the static demonstration, both running in the browser without a server; each caption links to the exact view captured. Base map © OpenStreetMap contributors (ODbL).

anomaly labels, the system computes statistical indicators over the retrieved records instead of training a model. The procedure runs unsupervised on the frontend. Consumption is aggregated by neighbourhood, each neighbourhood is compared against the others of its own borough, and the units with the largest deviation are ranked as candidates for inspection. Flagging patterns that require later review keeps the practical value of the component without presenting a prediction as if it had been trained against ground truth.

Formally, each neighbourhood g receives an atypicality score:

$$A_g = \left| \frac{x_g - \mu}{\sigma + \epsilon} \right|, \tag{1}$$

where x_g is the aggregate consumption of a neighbourhood, μ and σ are the mean and standard deviation of the neighbourhoods of its own borough —the comparison set implemented in `assets/atipicos.js`—, and ϵ prevents division by zero. The groups with the largest A_g are reported in the interface as zones with atypical consumption.

Two clarifications delimit the module. It is *exploratory* and not a causal predictor: a high A_g states that a zone departs from its comparison set, not why. And it does *not* perform real-time leak detection, since no leak reports, work orders or network telemetry are integrated into the warehouse and the evaluation in Section 5 rests exclusively on synthetically injected perturbations. Its role is the ranking function of the retrieval framework: the choropleth answers "how much is consumed here", the module answers "which units should I look at first", which is the ordering problem at the centre of information retrieval [7].

4 From the Dimensional Model to a Knowledge Graph

4.1 Rationale: Why a Star Schema, and What a Graph Adds

The star schema was not chosen by default. The dominant workload is the repeated aggregation of one numeric measure over three conformed dimensions with filters chosen interactively, which is what dimensional modelling was designed for [1]: constant join depth and aggregation by a single grouped scan. A graph would pay a traversal cost for joins that foreign keys already resolve. What a graph adds is complementary: *interoperability*, since a standard vocabulary makes the resource linkable to other municipal datasets without prior schema agreement, the recurring obstacle in the open-data literature [5]; *topological expressiveness*, since questions such as "which neighbourhoods adjacent to an atypical zone also exceed the borough mean" are natural traversals and awkward over a flat territorial dimension; and *entity reconciliation*, since stable identifiers would resolve the label heterogeneity our ETL absorbs through normalisation rules —the `w3id` namespace used here is not yet registered for dereferencing, which we leave as future work. We therefore keep the star schema as the analytical store and derive the RDF layer by a declarative mapping, so that neither representation is maintained by hand.

4.2 Vocabulary Alignment and R2RML Mapping

The correspondence between a star schema and the RDF Data Cube [12] is close to mechanical, which is what makes this layer inexpensive to maintain: a fact row becomes a `qb:Observation`, the measures become `qb:MeasureProperty`, the time and location dimensions become `qb:DimensionProperty`, the neighbourhood entity becomes a `geo:Feature` with its `geo:asWKT` geometry, territorial adjacency becomes the project-defined `agua:nearbyWithin1500m`, and the development index becomes a `skos:Concept` in a declared `skos:ConceptScheme` with an explicit numeric rank. The mapping is expressed in R2RML [13] and distributed with the archived release, declaring seven triples maps over both fact tables, the territorial dimension and its geometry, the derived adjacency, the temporal dimension and the index scheme. A companion `schema.ttl` supplies what the mapping alone cannot: the `qb:DataStructureDefinition` for both cubes, its components typed as dimension, measure or attribute, the cubic-metre unit and the ordinal scale. Loaded together with the browser graph, it satisfies the Data Cube integrity constraints IC-1, IC-2, IC-3 and IC-11 with zero violations, reproducible with `scripts/validar_datacube.py`; the check is stated for the published demonstration graph, not for the complete materialisation. The alignment is therefore a validated property rather than a nominal reuse of two terms. Materialising it yields on the order of 7.6×10^5 triples over 71,067 observations, 1,553 territorial features, 181 temporal entities and the four index members.

Two properties of the geospatial layer bound what the traversal can claim. The warehouse stores *points*, not polygons: each neighbourhood is the centroid of its readings, recovered from coordinates present in the source export but dropped by the original loading schema. The substitution is defensible at this scale —the 22,930 distinct coordinates are never shared between two neighbourhoods and the median neighbourhood diameter is 0.63 km— though coarse for the ten neighbourhoods above three kilometres. Because centroids are points, no Simple Features boundary relation applies: `geo:sfTouches` is undefined for point/point geometries, and relabelling a distance threshold as a standardised topological predicate would misstate what the data support. The mapping therefore declares its own `agua:nearbyWithin1500m`, documented in `schema.ttl` as centroid proximity below 1.5 km. Official polygons would allow `geo:sfTouches` in the strict sense; that is a substitution in the source data, and we leave it as future work.

4.3 Territorial Retrieval as Graph Traversal

The value of the layer is that it makes a class of questions expressible that the tabular interface does not naturally support. GeoSPARQL supplies the geometry model [14]; Listing 1.1 is the query the published interface executes verbatim, retrieving the neighbourhoods near a selected zone together with their own consumption, a single traversal that the relational interface resolves only through an ad-hoc spatial self-join, and which Fig. 1b shows executing in the published demonstration.

Listing 1.1: Territorial traversal over the knowledge graph, as executed by `vecindad.html`.

```
PREFIX agua: <https://w3id.org/cdmx/agua/>
PREFIX col:  <https://w3id.org/cdmx/agua/colonia/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?vecina ?alcaldia (ROUND(SUM(?c)) AS ?consumo) WHERE {
  col:465 agua:nearbyWithin1500m ?v .
  ?v rdfs:label ?vecina ; agua:borough ?alcaldiaIri .
  ?alcaldiaIri rdfs:label ?alcaldia .
  ?o agua:location ?v ; agua:totalConsumption ?c .
} GROUP BY ?vecina ?alcaldia ORDER BY DESC(?consumo)
```

That the graph is queryable, and not merely declarable, is shown without any server: the static deployment ships a browser-sized subset of 140,499 triples with a SPARQL 1.1 engine vendored in the release itself, so the demonstration depends on no external service. The subset is an aggregate, not a slice: its observations live at the neighbourhood-period grain in a separate `agua:aggobs/` namespace under `agua:consumoCDMXDemo`, linked to the complete dataset through PROV-O. Distinct identities avoid the contradictory RDF that reusing an IRI across grains would produce if the graphs were merged; the price is that a query written against the subset is not guaranteed to run unchanged against the full materialisation. The two representations agree where they overlap: aggregating consumption by bimester through SPARQL returns the same totals the reproduction script obtains in SQL.

Table 2: Data volume stored in the data warehouse

Table	Records
fact_consumo_agua	70,886
dim_ubicacion	1,553
dim_tiempo	181
dim_indice_des	4

5 Results and Evaluation

5.1 Integrated Data Volume

The ETL process consolidated the water-consumption information of Mexico City into the dimensional warehouse. Table 2 presents the stored volume. Each cardinality answers to a different grain and this must be read carefully. The 1,553 locations are the distinct borough–neighbourhood pairs and are coarser than the fact grain. The four index levels are the four members of the SACMEX ordinal scale, not three levels plus a residual. And the 181 time records are *not* reading dates: they are the daily observations of the climate series integrated alongside the consumption data, from which the dimension is populated. The consumption source carries three reading dates only, one per bimester, so the fact table distributes its rows over those three while the climate facts use all 181. The shared temporal dimension at daily grain is what allows both to be queried together.

5.2 Findings of the Consumption Analysis

The analysis of the 70,886 consolidated records reveals the following consumption patterns in Mexico City:

- **Territorial concentration:** The boroughs with the highest total consumption are Cuauhtémoc and Miguel Hidalgo, followed by Benito Juárez and Gustavo A. Madero (Table 3).
- **Temporal distribution:** Consumption shows variations across the bimesters available in the warehouse. In the aggregated query by bimester, the highest total consumption was observed in bimester 2, followed by bimester 3 and bimester 1. Therefore, no direct causal relationship with dry or rainy periods is established without integrating additional climatic variables.
- **Relationship with the development index:** Neighborhoods with a High development index show a higher average consumption than those with a Low index, a descriptive association discussed in Section 7.

Table 3: Five boroughs with the highest total water consumption

Borough	**Total consumption**
Cuauhtémoc	18 401 765.79
Miguel Hidalgo	14 490 397.15
Benito Juárez	13 780 118.64
Gustavo A. Madero	13 480 100.57
Álvaro Obregón	9 608 545.20

Table 4: Detection of injected anomalies, mean ± standard deviation over 20 seeds at equalised operating points. Reproducible with `evaluacion-ml/eval_anomalias.py`, a script reconstructed for this artefact.

Method	**P = R = F1**	**ROC-AUC**	**PR-AUC**
A_g (deployed)	0.581 ± 0.040	0.966 ± 0.013	0.677 ± 0.047
Isolation Forest	0.650 ± 0.053	0.971 ± 0.013	0.728 ± 0.058
LOF	0.493 ± 0.075	0.846 ± 0.036	0.464 ± 0.075

5.3 Quantitative Evaluation of the Atypical-Consumption Module

With no anomaly labels in the open source, the module is evaluated against injected ground truth. The protocol scores *the deployed function itself*, not a proxy: consumption is aggregated by neighbourhood, μ and σ are computed over the neighbourhoods of each borough, and A_g ranks them, which is exactly what `assets/atipicos.js` computes. The synthetic set reproduces the structure of the warehouse —1,553 borough–neighbourhood pairs over three bimesters— with 47 injected anomalies per run, and the experiment is repeated over 20 seeds. The three detectors are compared at equalised operating points, each emitting as many alerts as there are injected anomalies, so precision, recall and F1 coincide by construction and the areas under the curves characterise the ranking independently of any threshold (Table 4).

The deployed statistic is competitive with the strongest baseline: its ROC-AUC of 0.966 lies within one standard deviation of the 0.971 of Isolation Forest, and it recovers 58 % of the injected anomalies against 65 %. That 0.069 difference is modest relative to the combined between-seed variability, although Isolation Forest remains consistently superior at the evaluated operating point. LOF is clearly worse (0.846), since it looks for locally sparse neighbourhoods whereas the injected perturbations are departures in global magnitude. An interpretable borough-relative z-score therefore loses little ranking quality against an ensemble method a municipal analyst could not inspect, which supports A_g as a conservative exploratory filter, one that ranks a short list a utility can afford to inspect, and point to Isolation Forest as the natural improvement.

Two limitations remain. The perturbations are synthetic and their magnitude is chosen, so the figures measure sensitivity to known deviations rather than field performance against real leaks, for which the system is not designed and

Table 5: Response time of the retrieval workloads over the complete warehouse (70,886 fact rows), 200 executions after ten warm-up calls.

Workload	Mean (ms)	p95 (ms)	Executions
Paginated retrieval (`limit=100`)	32.3	35.2	200
Full aggregation by borough	24.1	27.0	200
Aggregation by neighbourhood and bimester	101.3	125.4	200
Top-10 neighbourhoods with ordering	23.1	23.6	200

no labelled data exist. And the comparison rests on a single injection model; a different anomaly shape could reorder the methods.

5.4 Performance Metrics

Lighthouse 13.4.1, run against the permanent static deployment, scores 89 on emulated desktop (first contentful paint 0.2 s, largest 0.3 s, total blocking time 200 ms) and 76 on emulated mobile over throttled 4G; accessibility, best practices and SEO score 100 on both. The remaining bound is structural: the static version downloads the full 3.9 MB export and aggregates on the client, so what a server computes once is computed by each visitor.

Query Performance over the Total Data Volume. Frontend metrics do not characterise the retrieval layer, so four analytical workloads were timed directly against the warehouse over all 70,886 fact rows: a paginated retrieval, a full-scan aggregation by borough, a full-scan aggregation by neighbourhood and bimester —the heaviest query the dashboard issues— and a top-N ranking. Each ran 200 times after ten warm-up calls on twelve logical cores with 32 GB of RAM and PostgreSQL 16.14 in a container, which is what makes the reported percentile meaningful. These are SQL latencies measured at the database, excluding serialisation and transport.

The profile, reported in Table 5, is the expected one for a star schema of this size: the three workloads that resolve through the dimensions stay near 25 ms while scanning the full fact table, whereas the aggregation by neighbourhood and bimester costs four times more because it groups over 1,553 locations rather than sixteen. The heaviest interactive query returns in about a tenth of a second, and the gap between mean and p95 stays below 25 % throughout, indicating that the cost is dominated by the scan rather than by contention.

5.5 Functional Tests

The operation of the ETL processes, the persistence in PostgreSQL, the parameterized queries and the information retrieval was verified. The tests included queries parameterized by year, bimester, borough, and neighborhood; aggregated retrieval (total and average consumption by neighborhood); generation of JSON responses; geospatial retrieval for visualization; and pagination with configurable limits. All modules produced results consistent with the expected structure. The dimensional model thus retrieves territorial information through combined filters of time, location and development index over the 70,886 records, supporting consumption queries by neighborhood and borough, temporal aggregation, coordinate retrieval, geospatial exploration and ordering by total consumption.

6 Discussion

Against the closest antecedents the contribution is one of *provenance and preservation* rather than of scale. Soares et al. [3] and Abecker et al. [4] begin from data an organisation already controls; this work begins from a public export with no schema, versioning or update cadence, which moves the difficulty from modelling to consolidation and is why the cleaning criteria and their yield are reported explicitly. The second difference is durability: those systems depend on a running service, whereas here interface, graph and data are preserved as static artefacts any reader can re-execute [8,9].

The practical implication is a caution about how the totals should be read. The boroughs that consume most are not those that consume most per inhabitant: Cuauhtémoc and Miguel Hidalgo concentrate commercial activity, and the warehouse supports the distinction because the source separates domestic from non-domestic billing —the non-domestic share reaches 32.5 % in Cuauhtémoc against 20.3 % in Gustavo A. Madero, twelve points apart between boroughs whose totals differ by barely a third. Total consumption is therefore not a per-capita indicator and should not be used as one for allocating supply interventions; the per-account average, also stored, is the appropriate measure.

7 Conclusions and Future Work

7.1 Conclusions

A data warehouse was presented for the territorial information retrieval of water consumption in Mexico City, built from open sources through a reproducible ETL process and published with a continuous-deployment methodology that guarantees a permanent functional demonstration. The solution consolidated 70,886 consumption records and 1,553 locations across the 16 boroughs into a dimensional star schema on PostgreSQL.

The findings reveal that water consumption in Mexico City is concentrated in the boroughs of Cuauhtémoc and Miguel Hidalgo (Table 3), and that it varies across the available bimesters without establishing in this work a causal relationship with dry or rainy periods.

The three central contributions —the consolidation ETL, the validated graph layer derived from it, and the always-accessible demo methodology— constitute a replicable method for bringing heterogeneous official open data to queryable and verifiable territorial information.

7.2 Threats to Validity and Limitations

Four limitations bound what may be concluded. The temporal extent is the first: the source publishes three bimesters of a single year, so the seasonal variation observed is a within-semester comparison and the climate correlation rests on three aggregated points, enough to describe a direction but not to estimate one. The second concerns the anomaly module, whose evaluation uses synthetic perturbations aligned with the detectors; its figures are an upper bound and say nothing about real leaks. The third is geometric: neighbourhoods are represented by centroids rather than polygons, so the adjacency projected as `agua:nearbyWithin1500m` is proximity within 1.5 km and not shared boundaries. The fourth is descriptive: the association between development index and consumption is an observation over the warehouse, and attributing it to a socioeconomic mechanism would require an external source the warehouse does not integrate.

7.3 Future Work

The most immediate line is to materialise the knowledge graph into a persistent triple store with a public SPARQL endpoint, and to replace the centroid-based adjacency with official neighbourhood polygons, which would make the topological predicates exact. Aligning the territorial identifiers with external authorities would let the resource be joined with other municipal datasets without prior schema agreement, which is the interoperability argument that motivates the graph layer. The anomaly module should be re-evaluated when labelled leak data exist, reporting complete precision–recall curves instead of a single operating point. Finally, the method transfers to other open urban-data domains —mobility, air quality and seismic risk— where the same combination of dispersed official files and absent query tooling recurs.

Declaration on the Use of Generative AI

The authors used a generative AI assistant, under their direction, for language editing, LaTeX formatting, drafting text subsequently revised and approved by the authors, and writing the auxiliary scripts distributed with the artefact. The research questions, the dimensional design, the ETL, the data collection, the experiments and the interpretation of the results are the authors' own work. Every reported quantity was recomputed from the two published source files and verified by the authors, and the scripts that perform that recomputation are distributed with the release. The authors take full responsibility for the content.

Disclosure of Interests

The authors declare that they have no conflicts of interest relevant to the content of this article.

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